

Help Box

Term	Simple Meaning	Prompts/Suggested starters
Research question	What are we trying to find out?	- How does _____ affect _____? - What is the effect of _____ on _____?
Hypothesis	What do you think will happen and why?	If _____, then _____ because _____.
Independent Variable	What will you change?	The thing I will change is _____.
Dependent Variable	What will you measure or observe?	The thing I will measure/observe is _____.
Control Variable	What will stay the same?	I will keep _____ the same. Other things that should stay the same are _____

Scenario/Prompt	Research Question	Hypothesis	Scientific reasoning	Independent variable	Dependent Variable	Control Variable
A student makes paper airplanes of different sizes and flies them to see which one travels the						

farthest						
Three identical plants are given different amounts of water every day for two weeks.						
A toy car is rolled down ramps of different heights. The distance travelled is measured.						
An ice cube is placed in sunlight and another ice cube is placed in the shade. The melting time is recorded.						
Sugar is added to hot water, warm water, and cold water. The time taken to dissolve is						

measured.						
A ball is rolled on different surfaces such as carpet, tiles, and grass.						
A flashlight is placed at different distances from a wall. The brightness of the light spot is observed.						
Different-sized parachutes are dropped from the same height. The time taken to reach the ground is recorded.						
Students sleep for different numbers of hours before taking a memory test.						

A student listens to music at different volume levels while studying and then takes a quiz.						
---	--	--	--	--	--	--